Crowdfunding Application

To run the application, click on https://lachezartsk.github.io/Crowdfunding

The project presents the interface for online donation via credit or debit card for crowdfunding campaigns. There are four projects in the presentation but the number of projects can be increased or decreased as needed.

The checks of the credit / debit card details are via validateCardForType from react-svg-credit-card-payment-icons as well as via a simulated database that contains details of valid cards, which is supposed to be supplied by the card issuing bank. Four types of cards are accepted: Visa, AmericanExpress, Mastercard, DinersClub.

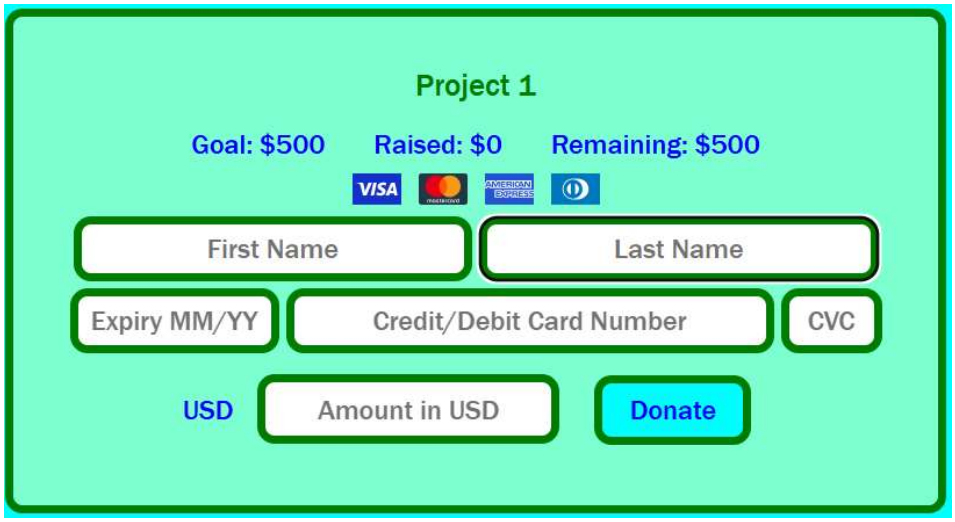
The check via ValidateCardForType matches the card number with the card type.
The check via the database tries to match all the card details: Name, Card Number, Expiration Date, CVC.
The transaction is allowed to proceed only if all checks are successful and the entered amount is greater than zero.

In the fields for card date, number, CVC and amount to donate, only digits are allowed to be entered.

Card date is automatically formatted in "MM/YY".
Card number is automatically formatted in "xxxx xxxx xxxx xxxx".
Card CVC is not formatted, since it consists only of three or four digits.
Amount to donate is automatically formatted in "xxx xxx xxx xxx".

Here is a quick walk through for one project and what would be displayed on the screen in different scenarios.

1. Start



2. Upon entering a valid card number that is from the accepted card type, a small card icon is displayed next to field CVC. This is regardless of whether all other card details are valid, which are checked upon clicking the button donate.

3. If all the card details are correct and the entered amount is greater than zero, upon clicking donate, the following is displayed. Goal, Raised and Remaining are updated upon each successful donation. And a popup message of gratitude is displayed for 4 seconds.

4. If the goal amount is reached, the button donate is disabled. There are two possibilities in regard to the total donated amount.

4.1 The donated amount matches exactly the goal amount. Then a message of gratitude is displayed, mentioning that the goal amount is reached. This is not a popup message and it is displayed till the project is removed.

4.2 The donated amount exceeds the goal amount. Then a message of gratitude is displayed, mentioning the value with which the goal amount is exceeded. This is not a popup message and it is displayed till the project is removed.

5. If any of the card details are not correct, upon clicking the button donate, a popup message about this is displayed for 4 seconds.

6. If the entered amount is not greater than zero, upon clicking the button donate, a popup message about this is displayed for 4 seconds.

If the goals for all four projects are reached, here is a sample of what the screen would be like.

The simulated database with valid card details has four sample entries. In order for the donation to be successful, the entry for the card must be one of the following.

Valid Card Sample One:
Number: "4242424242424242"
First Name: "John"
Last Name: "Smith"
Type: "Visa"
Expiry Date: "10/28"
CVC: "456"

Valid Card Sample Two:
Number: "378282246310005"
First Name: "John"
Last Name: "Adams"
Type: "AmericanExpress"
Expiry Date: "12/29"
CVC: "5678"

Valid Card Sample Three:
Number:"5555555555554444"
First Name: "Thomas"
Last Name: "Jefferson"
Type: "Mastercard"
Date: "08/29"
CVC: "234"

Valid Card Sample Four:
Number:"3056930009020004"
First Name: "Benjamin"
Last Name: "Franklin"
Type: "DinersClub"
Expiry Date: "05/28"
CVC: "248"

Project structure overview

Component CrowdfundingApp.jsx is the main entry point to run the application that is supplied to App.jsx The names of the folders, files, fields, methods, etc. are self-explanatory. Where necessary, comments are also provided.